

Bing Gao

Computational Mechanics Engineering Student / Fluid Simulation, Thermal Systems & Engineering Software

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Engineer with nearly 7 years in mobile software development and technical leadership, now applying experience from a product serving more than 2 million users to fluid simulation, finite-element analysis, thermal modelling, and numerical verification while completing a French engineering degree.

Nearly 7 years Software delivery and technical leadership	2M+ Users reached by a product built from scratch	2027 Engineering degree expected
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01

Experience

Engineering and product delivery

Aerodynamics Engineer

Vinci Eco Drive, ESILV Formula Student, Paris, France

- Led thermal-system modelling for the battery, inverter, and motor, coupling hydraulic, radiator, and transient coolant models to screen the E3 cooling architecture before procurement.
- Developed cooling-duct concepts within the aerodynamic package and used an OpenFOAM fan/core surrogate to qualify the numerical method before detailed vehicle fluid simulation.

Sep 2025 – Present

3D Modelling Intern

Felisiak Engineering & Development, Paris, France

- Reconstructed the ESA Space Rider from public blueprints for a launch-campaign review application and delivered separate review and animation-ready assets for Unity and Blender.
- Built a scripted Blender audit, rollback, and export workflow and prepared the engineering report, defense poster, and browser-ready GLB.

Apr – Aug 2026, 17 weeks

- Led Lao You from inception to more than 2 million users; built the product from scratch, wrote 70% of its initial core code, and helped establish it as a revenue source.
- Independently developed voiceprint-recognition algorithms and real-time gift-rendering effects for live broadcasts.
- Designed a Ruby-based delivery pipeline that automated testing, packaging, and distribution across Android and iOS.
- Implemented code and resource obfuscation plus certificate pinning to increase resistance to reverse engineering and traffic interception.
- Integrated React Native and built a hotfix mechanism for controlled runtime updates.

02

Education

Mechanics, computation, and software

Engineering Degree, Modelling & Computational Mechanics

Sep 2025 – 2027 expected

ESILV , Paris, France

- Computational fluid dynamics and finite-volume methods, finite-element methods, vehicle dynamics and powertrains, sustainable automotive engineering, structural optimization, machine learning, and digital twins.

Bachelor of Computer Science

Sep 2012 – Jul 2016, GPA 3.33/4.00

Changchun University of Science and Technology , Changchun, China

- Patent: “Mouse with gamepad function” (CN203870574U, issued Oct 8, 2014).
- First Provincial Prize, China Undergraduate Mathematical Contest in Modeling (2014).
- First Prize, 8th CUST ACM Programming Competition (2013).
- National Scholarship, Ministry of Education award, top 0.2% nationwide (2014–2015).
- First Class Scholarship, awarded five times for academic excellence (2012–2015).

03

Core skills

Tools and working languages

Simulation & numerical methods	OpenFOAM, Ansys Fluent, Ansys Mechanical, Star-CCM+, Abaqus, finite-volume fluid simulation, RANS, finite-element analysis, lattice-Boltzmann methods, POD/DMD, thermal-fluid modelling, mesh studies, and convergence studies.
CAD & 3D	CATIA, SolidWorks, and Blender.
Programming	Java/Kotlin, Python, Ruby/Ruby on Rails, C++, C, Rust, TypeScript/React, Node.js, and MATLAB/Simulink.
Languages	French (DELFI B2), English (C1), and Chinese (native).

Community

Developer education at scale

Google Developer Group Beijing , Beijing, China

- Organized free technical events that regularly drew more than 200 attendees and featured speakers from Google and Silicon Valley.
- Led technical knowledge sharing across mobile development, open-source tools, and emerging engineering topics.

Detailed methods, figures, and evidence records are available in the engineering portfolio.

Bing Gao, Paris